

ALM-II.

1) How do eigen values determine whether a system is stable or unstable?

Eigen values	stability
① Real & +ve	unstable
② Real & -ve	stable
③ Real & +ve, -ve	unstable
④ Real with	unstable

2) How eigen values related to the trace of a matrix?

→ The sum of the eigen values of a square matrix is equal to trace of the matrix

3) What can you say about eigen values of an upper triangular matrix?

→ The eigen values of an upper triangular matrix are exactly its diagonal entries

4) If all eigen values of a matrix are +ve, what can you conclude about the system stability?

→ If all eigen values are +ve
System is unstable

5) Determine the eigen values & Eigen vectors of the matrix,

→ Given $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

By char eqn find eigen values $[IA - \lambda I]$
char matrix $[A - \lambda I]$

$$\begin{bmatrix} 6 & -2 & 2 \\ -2 & -3 & -1 \\ 2 & -1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix}$$